Sparse Dictionary Learning for Interpretable Lung Disease Diagnosis from 3D Chest CT

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Abstract

Lung disease diagnosis from 3D chest CT scans faces critical challenges in computational efficiency and clinical interpretability. Traditional 3D Convolutional Neural Networks require extensive labeled data and GPU resources, while post-hoc explainability methods like Grad-CAM provide insufficient granularity for medical decision-making. While deep learning models achieve high accuracy, their black-box nature and substantial computational demands limit clinical adoption. This project proposes a sparse dictionary learning framework that addresses these limitations through inherently interpretable feature representations. The methodology employs a two-phase frozen dictionary learning approach combined with Label-Consistent K-SVD (LC-KSVD) to capture both normal anatomical patterns and disease-specific pathological features. Dictionary atoms are mapped to clinically meaningful structures such as Ground-Glass Opacity (GGO) and consolidation, enabling transparent diagnostic reasoning. Each dictionary atom corresponds to specific volumetric patterns, allowing clinicians to trace classification decisions back to anatomical features rather than relying on post-hoc visualization techniques. Building on this inherently interpretable representation, volumetric quantification using Hounsfield Unit (HU)-based segmentation and Consolidation-to-GGO (C/G) ratio calculation integrates domain-specific clinical metrics into the classification pipeline. The proposed system will be benchmarked against standard K-SVD and 3D CNN baselines using CT-RATE and ReXGroundingCT datasets. The sparse representation framework naturally handles class imbalance and requires fewer training samples than deep learning approaches, making it particularly suited for rare pathologies. This research aims to demonstrate that sparse representation methods can provide clinically interpretable lung disease classification while maintaining competitive performance with reduced computational overhead.

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Acronyms

AI Artificial Intelligence

C/G Consolidation-to-Ground-Glass-Opacity

CNN Convolutional Neural Network

CT Computed Tomography

GGO Ground-Glass Opacity

HU Hounsfield Unit

KSVD K-Singular Value Decomposition

LC-KSVD Label-Consistent K-Singular Value Decomposition

XAI Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality[1]. In the diagnosis of lung disease, assessing the pattern, location, and regional distribution of involvement is the domain of radiology. Chest X-rays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, their diagnostic sensitivity is limited, as they capture only a two-dimensional image of the chest. This causes anatomical superposition, where overlapping tissues and organs are projected onto a single plane, allowing visualization of only about 70% of the lung volume[2].

Computed Tomography (CT) is a radiographic technology that combines X-ray technology with computer reconstruction to produce images of cross sections of the human body[3]. This overcomes many of the limitations of radiography by eliminating anatomical superposition and enabling full volumetric assessment of the lungs. CT produces high-resolution slices that reveal fine structural details not visible in standard chest radiographs.

A defining characteristic of a CT scan is the Hounsfield Unit (HU), a quantitative scale of radiodensity that allows tissue properties to be measured. In pulmonary imaging, HU values allow for the objective differentiation of tissues such as normal parenchyma, Ground-Glass Opacity (GGO), and consolidation that might appear visually similar. GGO appears as hazy increased opacity of lung, with preservation of bronchial and vascular margins. Consolidation is more opaque, and appears as a homogeneous increase in pulmonary parenchymal attenuation that obscures the margins of vessels and airway walls[4]. While GGO often suggests a reversible or early-stage condition, consolidation usually indicates a more aggressive or advanced stage. The Consolidation-to-GGO (C/G) ratio is a quantitative metric calculated by dividing the volume or area of consolidation by the volume or area of GGO within a lesion or throughout the lungs.

The traditional approach for CT analysis relies on a "slice-by-slice" approach where a radiologist views a series of axial cross-sections and identifies the index slice (the single image where a pathology appears at its largest diameter)[5]. A 2D slice only captures a small fraction of a 3D structure. If a nodule is irregularly shaped, the largest slice may not represent the true extent of the disease. Studies show that 2D measurements are highly sensitive to the patient's orientation and the specific slice chosen, leading to high inter-observer variability. 2D models

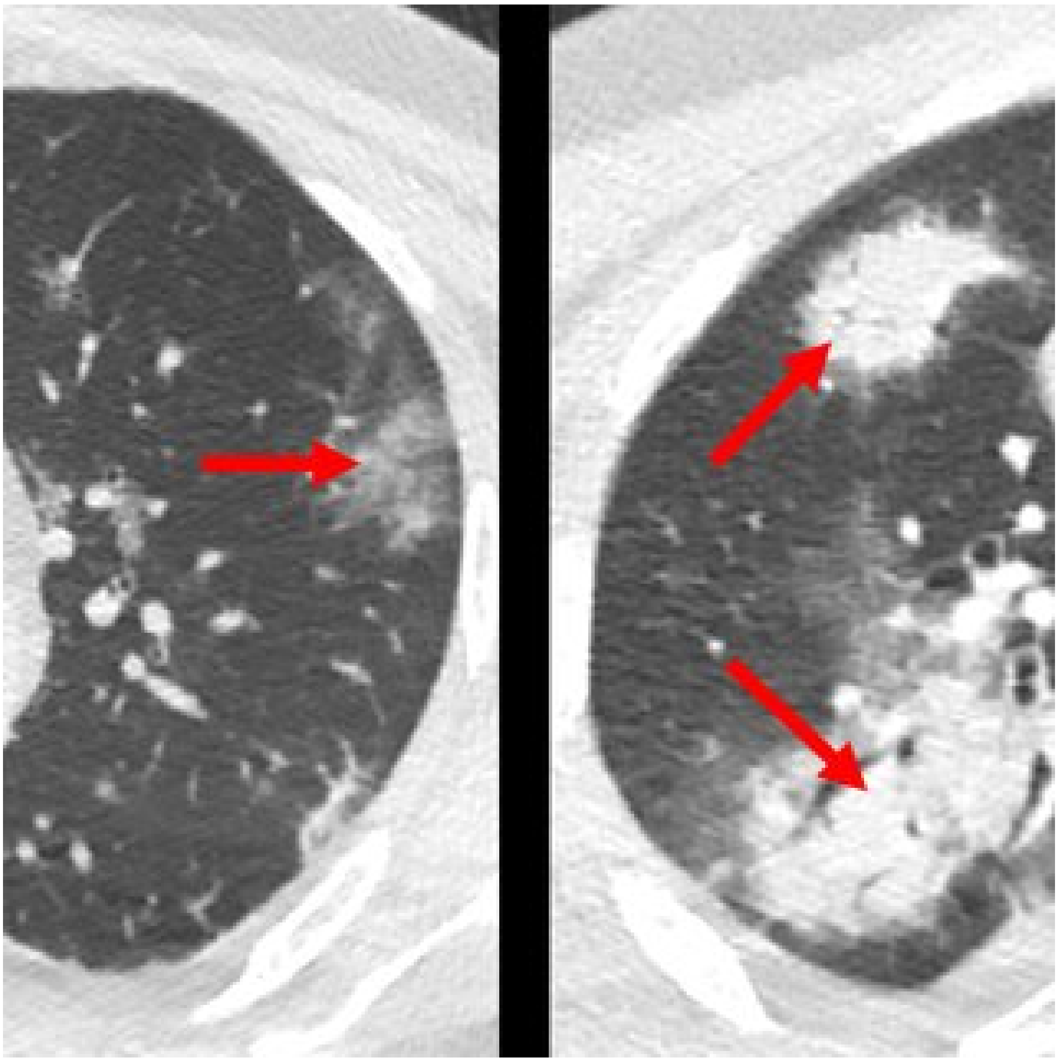

Figure 1.1: Examples of GGO and CONS on axial slices of lung CT scans from COVID-19 patients. Arrows point to the areas affected by the infection. Left: GGO Right: CONS.

also treat each slice as an independent data point, losing the "spatial continuity" between adjacent tissues.

3D volumetric analysis represents a quantitative shift wherein the entire dataset is used to reconstruct the lung's anatomy as a solid volume. This enables more accurate characterization of lesion size, shape, and spatial distribution, while reducing sensitivity to slice selection and viewing angle. 3D representations better capture disease morphology and progression, providing a more robust foundation for quantitative analysis and automated learning-based methods.

Modern radiology increasingly relies on Artificial Intelligence (AI) to streamline the acquisition of radiologist analyses on chest X-rays. However, the exponential increase in data volume generated by 3D CT scanners presents a profound challenge for human interpretation and computational analysis. While the prevailing research trend has focused heavily on the deployment of 3D deep learning architectures, such as Convolutional Neural Networks (CNNs), these models frequently encounter significant hurdles regarding computational overhead, data scarcity, and

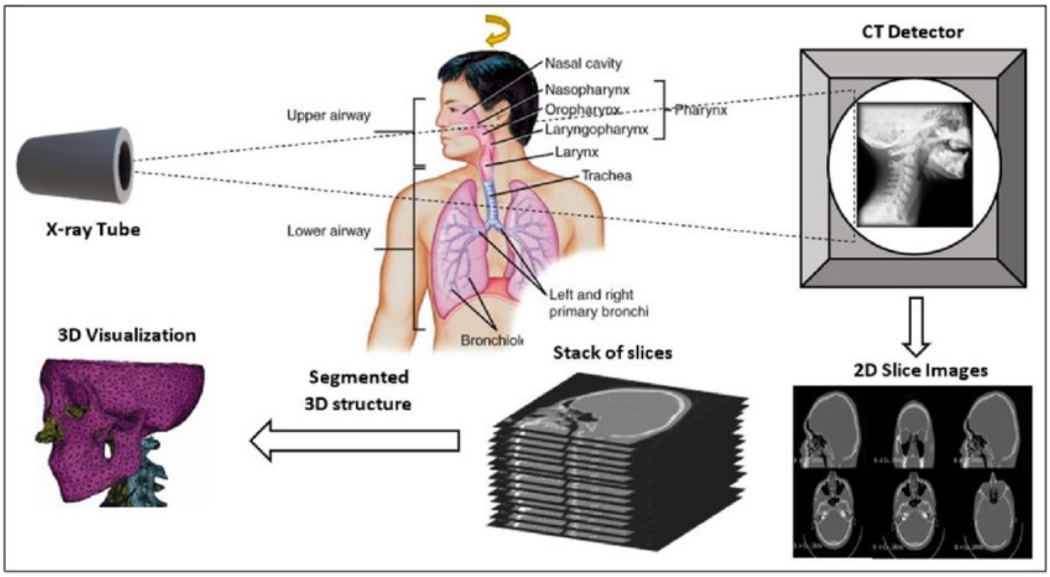

Figure 1.2: 3D Imaging and visualization process of Computed Tomography for the craniofacial region.

a critical lack of interpretability. The research proposed herein advocates for a departure from these "black box" paradigms, suggesting that sparse representation methods offer a more intutive framework for the volumetric identification of lung diseases by ensuring clinical interpretability.

1.2 Problem Statement

Modern radiology increasingly relies on AI to manage the vast quantities of 3D data produced during routine screenings. 3D deep learning has been widely applied to the automatic diagnosis of lung diseases, achieving remarkable success in tasks such as segmenting suspected infection regions and differentiating between various types of carcinoma. Despite these advancements, the inherent complexity of 3D CNNs introduces a set of constraints that are often overlooked in the pursuit of high accuracy scores.

1.2.1 Computational Overhead

Processing a 3D CT scan involves treating stacked 2D slices as a single volumetric matrix, which demands significantly more GPU memory and processing power than standard image analysis. Unlike ordinary RGB images, CT scans consist of single-channel data with varying voxel dimensions and resolution, which complicates spatial alignment and generalizability across different scanners. Furthermore, the "curse of dimensionality" remains a persistent threat; as the network depth increases to capture global features, the risk of gradient explosion and slow convergence becomes more pronounced.

1.2.2 Interpretability Bottleneck

Beyond computational limits, the clinical utility of deep learning is hampered by the interpretability paradox. While tools like Grad-CAM provide visual heatmaps to localize lesions, these post-hoc explanations are often insufficient for medical professionals who require a granular understanding of the model's decision-making process. In high-stakes environments, a model that shruggingly produces a diagnosis without an audit trail of its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance. This transparency gap has led to a renewed interest in "open book" models that are interpretable by design, where the internal logic is clear from the outset.

1.3 Objectives and Scope

1.3.1 Objectives

The primary goal of this project is to address the limitations of existing AI based lung disease diagnosis systems, specifically high computational overhead and lack of clinical interpretability, by developing an inherently interpretable and computationally efficient dictionary learning framework. The proposed approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facillitates validation and adoption in radiological workflows.

1. Develop an interpretable dictionary learning framework for lung disease classification from 3D chest CT scans
    - Implement a two phase training strategy to learn normal and pathological dictionary atoms using a publicly available annotated 3D chest CT dataset
    - Integrate label consistency to improve class discrimination while maintaining interpretability
2. Improve model interpretability through visualization of learned dictionary atoms
    - Visualize dictionary atoms in 3D and map them to anatomical and pathological lung features
    - Analyze and compare sparse coefficient patterns across different disease classes
3. Evaluate the effectiveness of the proposed framework
    - Benchmark performance against standard KSVD without label consistency
    - Compare results with a 3D CNN baseline (3D-ResNet)
4. Integrate clinically relevant volumetric features into the analysis

- Perform HU based segmentation to identify Ground-Glass Opacity (GGO) and consolidation
- Compute the Consolidation toGGO (C/G) ratio and relate model predictions to clinical thresholds

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning-based framework for lung disease classification using volumetric 3D chest CT data, with the following limitations:

- The proposed solution is limited to volumetric CT scans of the thorax and does not consider other imaging modalities such as chest X-rays, MRI, or PET scans.
- The methodology is restricted to dictionary learning-based approaches, with particular emphasis on KSVD and LC-KSVD End-to-end deep dearning models such as CNNs are not the primary focus of this work and are included only as baseline comparators for performance benchmarking.
- Clinically, the analysis is confined to the characterization of lung parenchymal abnormalities, specifically GGO and consolidation patterns. Quantitative metrics such as HU-based segmentation and the C/G ratio are employed to support volumetric analysis. The study does not address disease prognosis or treatment outcome prediction.
- The scope excludes real-time clinical deployment, cross-institutional generalization studies, and regulatory validation. The findings are intended to demonstrate the feasibility and interpretability of sparse representation learning for 3D lung disease analysis rather than to provide a clinically deployable diagnostic system.

Chapter 2

Literature Review

2.1 Previous Work

2.1.1 Deep Learning for Lung Disease Diagnosis

Numerous techiniques have emerged in the field of automated lung disease diagnosis, each attempting to balance accuracy with computational feasibility. Early approaches relied on manual feature engineering and classical classifiers like Support Vector Machines (SVM) and Random Forests.[6] While these methods provided strong interpretability, they often lacked the flexibility to handle the noise and variance found in CT data[7]. The emergence of deep learning fundamentally changed the landscape of CT analysis, especially in areas such as;

- Image Segmentation and Classification
- Advancing Diagnostics with AI and CAD Systems
- Prognostics with Radiomics and Predictive Analytics
- Workflow Optimisation Using AI

Convolutional Neural Networks (CNNs) have emerged as the dominant paradigm for automated analysis of CT scans[8]. Recent surveys indicate that deep learning models achieve classification accuracy exceeding 98% for multi-disease detection tasks[9], representing a signficant improvement over traditional machine learning approaches. Contemporary research has increasingly focused on 3D CNN architectures that treat CT scans as complete volumetric data rather than independent 2D slices. This volumetric approach captures spatial relationships within lung tissue more effectively, leading to improved sensitivity in detecting subtle pathological changes. Li et al. (2024)[10] demonstrated that preprocessing techniques such as lung region extraction, combined with advanced backbone architectures like ResNeSt50, significantly enhance classification performance for multiple lung pathologies [11].

2.1.2 Sparse Dictionary Learning in Medical Imaging

Sparse dictionary learning represents an unsupervised machine learning paradigm for extracting interpretable features from medical images. This technique learns a set of dictionary elements or basis vectors that represent data in a sparse manner. In the context of 3D CT, a volumetric patch $y \in \mathbb{R}$ can be approximated by $y \sim Dx$, where $D \in \mathbb{R}^{\ltimes\frown\mathbb{K}}$ is the dictionary and $x \in \mathbb{R}^{\rceil}$ is a sparse vector containing only a small number of non-zero coefficients (figure 2.1). The strength of dictionary learning lies in its ability to extract local, structural information from noisy medical images while requiring less labeled data than purely supervised deep learning approaches. This characteristic is particularly valuable for rare diseases with limited training samples[12].

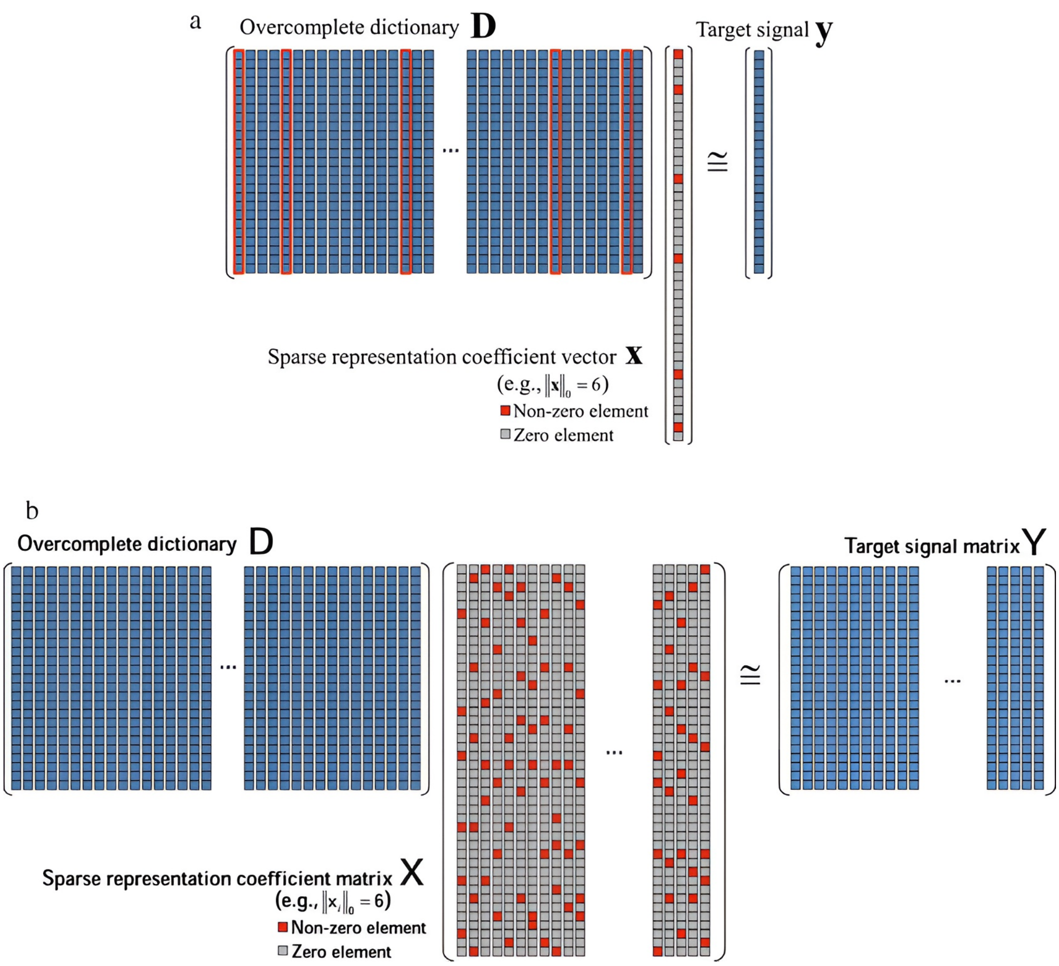

Figure 2.1: Brief overview of the sparse representation. (a) Sparse linear combination Dx approximating y, where $||x||_0 = 6$ and (b) two matrices D and X calculated in K-SVD algorithm 36 from Y, where D is a dictionary matrix, X is a matrix including representation coefficient vectors, and Y is a matrix including target signals.

The success of these models hinges on the quality of the learned dictionary. Traditional methods like KSVD focus primarily on minimizing the reconstruction error, which is effective for image denoising and super-resolution but lacks the discriminative power required for classification[13]. To address this, supervised dictionary learning techniques have emerged, which incorporate label information directly into the dictionary construction process. The LC-KSVD algorithm enhances the traditional KSVD by adding a label consistency constraint and a classification error term to the objective function[13]. This approach forces images from the same class to have similar sparse codes, thereby increasing the discriminative power of the model.

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature-oriented Dictionary Learning (DFDL) has shown the ability to learn class-specific dictionaries that achieve high classification accuracy even with limited training data—a critical advantage when generous training samples are not available[14]. In neuroimaging, multi-feature kernel supervised dictionary learning approaches have demonstrated superior performance in distinguishing Alzheimer's disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times[15].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dictionary learning and label-consistent LC-KSVD to classify COVID-19 CT scans, achieving 89% accuracy on the CC-CCII dataset[16]. The framework's key advantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature representations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease patterns on HRCT images, effectively handling the high variety and complexity of patterns[17].

2.1.3 Explainable AI in Medical Imaging

Clinical radiology requires an understandable method for determining how the disease occurs. The "black-box" nature of deep neural networks creates significant barriers for clinical adaptation. Physicians need to understand diagnostic reasoning to ensure accountability and patient safety. [18]. To address these challenges, a range of explainable artificial intelligence (XAI) techniques have been developed to provide insight into model decision-making. Predominant XAI approaches in medical imaging include:

Perturbation-Based Methods: These techniques evaluate feature importance by systematically altering specific image regions and observing how these changes affect the model's output.

Gradient-Based Methods: These methods highlight image areas that most significantly utilizing backpropagation gradients influence a classification. Grad-CAM is particularly predominant because it is applicable to nearly any network architecture [19].

Decomposition-Based Methods: Techniques like Layer-Wise Relevance Propagation (LRP) distribute the output of the final layer back through the network recursively. This process calculates a relevance score for each neuron, generating a heatmap that provides a clearer distinction of feature contribution.

These explainability techniques greatly increase clinician comprehension and confidence in adapting AI-driven diagnostic results, making it easier to integrate with conventional radiological workflows while maintaining regulatory compliance [19].

2.1.4 Volumetric Clinical Feature Extraction

The core of pulmonary diagnostic analysis lies in the identification and quantification of specific radiographic patterns. GGO is a hallmark sign in chest CT, characterized by hazy increased attenuation that does not obscure the underlying vascular markings. It is a non-specific finding that can represent fluid, fibrosis, or partial collapse of alveoli, and it is particularly prevalent in the early stages of diseases like COVID-19 or certain types of non-small cell lung cancer.

Volumetric analysis allows for the quantification of these features beyond simple 2D identification. By segmenting the lung parenchyma and using HU thresholding, the percentage of lung tissue affected can be calculated by specific patterns. A critical metric in this domain is the C/G ratio. Clinical data shows that as GGO progresses to consolidation, the severity of the patient's condition increases[20].

2.2 Gaps in Literature

The proposed solution is designed to address several gaps in the current literature, as summarized in table 2.1:

Table 2.1: Gaps addressed by proposed approach vs existing solutions

Aspect	Existing Solutions	Proposed Solution
Interpretability	Post-hoc explanations (Grad-CAM, LIME, SHAP); Black-box features with no semantic meaning	Inherently interpretable dictionary atoms representing concrete anatomical structures (nodules, GGO, consolidations)
3D Volumetric Analysis	Limited integration of interpretability with full 3D data; mostly 2D slice-by-slice or post-hoc 3D explanations	Full 3D volumetric analysis with interpretable sparse representations
Feature Representation	Black-box features with no semantic meaning	Dictionary atoms represent concrete, visualizable anatomical structures (nodules, GGO, consolidations)
Label Consistency	Standard K-SVD focuses only on reconstruction error, lacks discriminative power	Dictionary learning with sparse representations is more efficient than deep 3D CNNs
Clinical Metrics Integration	Often limited to classification accuracy; lacks integration of domain-specific quantitative metrics	Integrates HU-based segmentation for GGO/ consolidation detection and C/G ratio calculation
Clinical Metrics Integration	Deep learning requires large labeled datasets	Dictionary learning is effective with limited labeled data; particularly valuable for rare diseases
Handling Class Imbalance	Requires synthetic data generation, complex loss weighting, or meta-learning	Sparse dictionary learning naturally handles few-shot scenarios for rare diseases

2.2.1 Inherently Interpretable Architectures

While 3D CNNs have demonstrated superior performance for lung disease classification from CT scans, the majority of explainability research has focused on 2D imaging modalities (chest X-rays) or individual CT slices analyzed independently [21]. There remains a significant gap in developing inherently interpretable feature representations specifically designed for full 3D volumetric data [22].

Modern XAI methodologies (e.g., 3D Grad-CAM) frequently utilize post-hoc explanations to visualize important regions but do not provide semantically meaningful feature representations that clinicians can directly relate to pathological patterns [22]. These approaches explain model decisions after they are made but fail to tackle their underlying black-box nature. The extension of 3D saliency maps to interpretable concepts such as specific lesion types, textural patterns, or anatomical landmarks remains largely unexplored. This gap is particularly significant, because pulmonary pathology often manifests as spatial patterns spanning multiple CT slices (e.g., centrilobular versus panlobular emphysema distributions).

Sparse dictionary learning presents an alternative approach where learned dictionary elements can represent interpretable features such as nodules, GGO, consolidations, etc [12]. Every classification decision links to particular, tangible dictionary components, offering inherent rather than post-implementation clarity. Despite the integration of sparse dictionary learning principles with contemporary 3D deep learning models, significant exploration in 3D CT analysis is still lacking [23]. The existing hybrid methods, such as deep dictionary learning[24], have mainly been applied to 2D natural images rather than to volumetric medical imaging.

2.2.2 Frozen Dictionary Learning for Pathological Differentiation

A primary innovation of the proposed system is the utilization of "frozen" dictionary learning. In conventional dictionary learning, all atoms are updated simultaneously, which can lead to a phenomenon where "normal" atoms inadvertently absorb pathological features [13], reducing the model's ability to clearly isolate anomalies. Frozen dictionary learning addresses this by employing a staged training strategy that partitions the dictionary into anatomically standard and pathologically specific components [25] .

The first phase involves learning a base dictionary using only healthy lung data [26]. These atoms capture the expected appearance of lung parenchyma, bronchi, and pulmonary vasculature. Once this base dictionary is optimized, it is "frozen" (its atoms are held constant and are no longer updated) [25]. In the second phase, additional dictionary atoms are introduced and trained specifically on anomalous data such as CT scans containing viral pneumonia or adenocarcinoma. Because the standard anatomical features are already accounted for by the frozen atoms, the new atoms are forced to capture only the unique variations or lesions that represent the disease state.

When a query 3D patch is analyzed, the resulting sparse coefficient vector reveals exactly which atoms [27]. If the representation relies heavily on the anomalous portion of the dictionary, the clinician can identify the specific type of pathology present by mapping those atoms back to their visual counterparts in the original signal domain. This "feature separation" ensures that the diagnostic process is transparent and verifiable, moving away from the hidden logic of deep neural networks.

2.2.3 Handling Class Imbalance with Interpretable Models

Rare lung pathologies such as certain cancer subtypes (e.g., large cell carcinoma, carcinoid tumors), uncommon ILD patterns (e.g., pleuroparenchymal fibroelastosis), and emerging infectious diseases—present severe class [9]. Deep learning research has addressed this through weighted loss functions, oversampling strategies, synthetic data generation (e.g., progressive growing GANs), and meta-learning approaches [28]. However, the application of class imbalance mitigation techniques within sparse representation frameworks for 3D CT analysis is underexplored [12]. Dictionary learning methods that can effectively learn representative atoms for rare categories while maintaining interpretability for common pathologies require further development [27]. The few-shot learning capabilities of sparse dictionary approaches offer potential advantages for rare disease applications, but have not been systematically evaluated in pulmonary imaging contexts.

Chapter 3

Methodology

The research consists of 7 main phases as shown in figure 3.1:

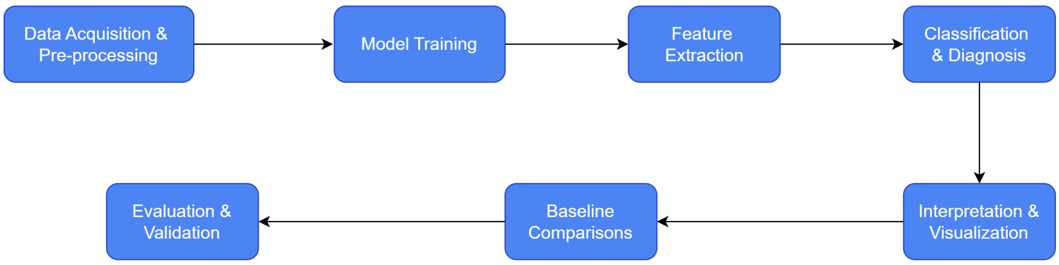

Figure 3.1: Overall Methodology

3.1 Data acquisition and pre-processing

We selected two publicly available 3D chest CT datasets, CT-RATE[29] and ReXGroundingCT[30], for training and evaluation due to their complementary nature. Their characteristics are summarized in table 3.1:

Table 3.1: Characteristics of CT-RATE and RexGroundingCT datasets

Feature	CT-RATE dataset	RexGroundingCT dataset
Type of Data	3D non-contrast chest CT volumes with corresponding radiology text reports and multi-abnormality labels	Subset of CT-RATE scans paired with free-text findings and pixel-level 3D segmentation masks
Number of Scans	25,692 non-contrast chest CT studies (50,188 volumes after reconstruction)	3,142 non-contrast chest CT scans
Number of patients	21,304 unique patients.	Subset selected from CT-RATE
Associated Reports	Full radiology reports (findings and impressions) with structured multi-abnormality labels (18 categories)	Standardized radiology reports from CT-RATE used to extract free-text findings
Annotations	Text-image pairs with structured labels; no voxel-wise ground truth segmentations	Manual pixel-level 3D segmentations of findings extracted from reports (8,028 findings; 16,301 entities)

Data pre-procesising corrections such as intensity normalization and spacing adjustments have already been implemented directly in the dataset. If further pre-processing is required, standard techniques such as resampling to a uniform voxel size, intensity normalization, and lung segmentation, shall be implemented before feeding the data into the model. Both datasets are released under MIT licenses that allow use for non-commercial academic research.

3.2 Model Training

Model training will be implemented using a two-phase frozen dictionary learning approach[31] to capture both normal and pathological patterns in chest CT scans. The dataset will be divided into train,test, and validation sets.

Phase 1: Learning the Normal Dictionary
A dictionary is trained using only healthy or normal samples, enabling the model to capture typical lung anatomy.

Phase 2: Learning Pathological Atoms
With the normal dictionary fixed, additional atoms are learned[32] to represent disease-specific pathological features while leveraging the frozen dictionary during sparse coding.

We extend the dictionary learning approach using LC-KSVD to improve the discriminative power of the learned sparse representations. The label consistency constraint will produce similar sparse codes for CT samples belonging to the same disease category. LC-KSVD will be formulated to jointly optimize three key objectives:

1. Accurate reconstruction of CT image patches using the learned dictionary.
2. Consistency of sparse representations within the same disease class.
3. Effective classification of lung conditions based on the sparse codes.

Appropriate weighting will be applied to balance these objectives and hyperparameters will be selected through validation experiments.

3.3 Classification and Disease Diagnosis

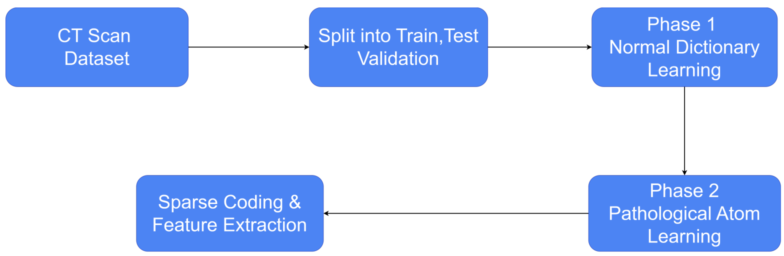

Figure 3.2: Model Training Workflow

SVM will be used as the final disease classification model due to the following reasons:

1. Ability to handle high-dimensional sparse features[33] produced by dictionary learning and LC-KSVD
2. Effectiveness when the number of features is much larger than the number of samples

3. Strong generalization due to the maximum-margin principle

SVMs are also known to be robust when dealing with limited and imbalanced medical data, making them a suitable choice for disease classification tasks.

The model will be implemented using the scikit-learn library in Python and will receive the sparse codes generated from the learned dictionary as input features. A linear kernel will be used as the initial choice due to its effectiveness with sparse representations. The classifier will predict multiple disease categories, with the number of output classes matching the labeling structure of the CT-RATE dataset.

3.4 Interpretability Analysis and Visualization

To improve interpretability and clinical relevance, the learned dictionary atoms will be visualized and analyzed through an interactive user interface utilizing 3D rendering techniques. Each dictionary atom will be mapped back to the original CT image space, allowing us to see which anatomical regions or pathological patterns the model has learned to represent. This mapping is guided by the ReXGroundingCT annotations, which provide precise spatial localization of abnormalities within the 3D imaging volume.

3.5 Volumetric Feature Extraction and Quantification

Volumetric quantification of pulmonary patterns will be performed using HU-based segmentation to differentiate between GGO and consolidation regions. GGO will be identified using HU thresholds in the range of -700 to -500 HU, where increased attenuation preserves underlying vascular structures. Consolidation will be segmented using thresholds between -500 and 100 HU, corresponding to homogeneous opacification that obscures bronchial and vascular margins. The segmented regions will be extracted from the 3D CT volumes to calculate the total volume of each pattern type.

The C/G ratio will be computed by dividing the total volume of consolidation by the total volume of GGO across the entire lung parenchyma. This quantitative metric provides an objective measure of disease severity and progression, with higher ratios indicating more advanced pathological states. The calculated C/G ratios will be compared against model predictions and clinical thresholds to validate the diagnostic utility of the learned dictionary representations. These volumetric features will complement the sparse coding analysis, enabling a more comprehensive evaluation of the model's ability to capture clinically relevant patterns.

3.6 Baseline Comparisons and Benchmarking

The proposed method will be compared against well established baseline approaches commonly used in medical imaging research. These baselines are chosen to represent both traditional machine learning techniques and modern deep learning models. Standard K-SVD is included as a baseline to highlight the advantage of incorporating label information through LC-KSVD. By comparing these two sparse representation methods, the contribution of label consistent dictionary learning to disease classification performance can be assessed.

In addition, several 3D CNN architectures will be used as deep learning baselines, as they are considered state-of-the-art for volumetric CT analysis. These models are capable of learning complex spatial patterns across multiple slices and therefore provide a strong benchmark against which the proposed method can be evaluated. Existing implementations of these networks will be adopted to ensure reliability and consistency. All models will be trained and tested using the same dataset, although training strategies may differ due to the higher data and computational demands of CNN-based approaches.

All methods will be implemented using suitable and widely adopted frameworks, including scikit-learn for classical machine learning and sparse coding methods, and PyTorch or TensorFlow for deep learning based models.

3.7 Evaluation Metrics and Validation Strategy

The performance of the proposed model will be evaluated using accuracy, precision, recall, F1-score, and AUC-ROC, which are standard and widely accepted metrics in medical image analysis. Greater emphasis will be placed on accuracy and recall, as they are especially important in a clinical context to ensure overall reliability and to minimize missed disease cases, while the remaining metrics provide additional insight into class-wise performance and discriminative ability.

Chapter 4

Timeline and Resources Required

4.1 Timeline

The project will be carried out within 2 semesters (24 weeks) according to the timeline given in table 4.1.

4.2 Resources Required

The project requires several Hardware and Software resources as summarized in table 4.2.

Table 4.1: Overall Project Time Plan

Project Task	**Semester 07**												**Semester 08**											
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12
Dataset acquisition & preprocessing																								
Implement frozen dictionary + LC-KSVD & initial experiments																								
Implement volumetric analysis & feature extraction																								
Clinical validation & comparative experiments																								
Final experiments & visualization																								

Table 4.2: Resources Required for the project

Resource Category	Specification
Hardware	
GPU	NVIDIA GPU (minimum 16GB VRAM) for training 3D CNN baselines and processing volumetric CT data
CPU	Multi-core processor (minimum 8 cores) for dictionary learning algorithms
RAM	Minimum 32GB for handling large 3D CT volumes in memory
Storage	Minimum 500GB for storing CT-RATE and ReXGroundingCT datasets plus processed data and model checkpoints
Software & Libraries	
ML Frameworks	PyTorch or TensorFlow for 3D CNNs scikit-learn for SVM classification
Medical Imaging	SimpleITK or NiBabel for CT processing PyDicom for DICOM handling scikit-image for preprocessing
Visualization	Three.js or VTK for 3D visualization ITK-SNAP or 3D Slicer for qualitative analysis
Datasets	
CT-RATE	50,188 3D chest CT volumes with radiology reports (publicly available under MIT license)
ReXGroundingCT	3,142 chest CT scans with annotated 3D segmentation masks (publicly available under MIT license)

Chapter 5

Conclusion

This proposal presents a sparse dictionary learning framework for interpretable lung disease diagnosis from 3D chest CT scans. By employing frozen dictionary learning with LC-KSVD, the proposed approach addresses fundamental limitations of existing deep learning methods: computational overhead, data scarcity, and lack of clinical interpretability. The two-phase training strategy ensures clear separation between normal anatomical features and pathological patterns, enabling transparent diagnostic reasoning that clinicians can verify and trust.

The integration of volumetric clinical metrics, specifically HU-based segmentation and C/G ratio calculation, bridges the gap between machine learning predictions and established radiological practice. Dictionary atoms provide semantically meaningful representations that map directly to observable pathological structures, moving beyond post-hoc explanations toward inherently interpretable models.

Benchmarking against K-SVD and 3D CNN baselines will validate the effectiveness of label-consistent dictionary learning for medical image classification. The use of publicly available datasets (CT-RATE and ReXGroundingCT) ensures reproducibility and facilitates future research extensions. Ultimately, this work contributes to the development of "open book" AI systems that prioritize clinical transparency alongside diagnostic performance, offering a viable alternative to black-box deep learning approaches in high-stakes medical applications.

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